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Advanced Macroeconomics I

Problem Notes and Selected Solutions for Romer 5ed

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Preface

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CHAPTER 1

PROBLEM 1.1 *Basic properties of growth rates.*

Use the fact that the growth rate of a variable equals the time derivative of its log to show:

(a) The growth rate of the product of two variables equals the sum of their growth rates. That is, if $Z(t) = X(t)\,Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)}$$

(b) The growth rate of the ratio of two variables equals the difference of their growth rates. That is, if $Z(t) = X(t)/Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)}$$

(c) If $Z(t) = a\,X(t)^{\alpha}$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \alpha\,\frac{\dot{X}(t)}{X(t)}$$

NOTE Because instantaneous growth g on continuous-time is naturally expressed in exponential form, $\lim_{n\to\infty}(1+g/n)^{nt} = e^{gt}$, the rules above reflect a fundamental duality: logarithms turn products into sums, while exponentials turn sums into products.

NOTE This problem gives four useful rules for growth rates. For products, add the growth rates. For ratios, subtract them. For powers, multiply the growth rate by the exponent. Multiplying a variable by a constant does not change its growth rate. These rules make growth-rate calculations much easier.

What about sums and differences, such as $(\dot{X \pm Y})$? They also have growth-rate formulas, but these formulas are less useful and are not needed here. See the *Advanced Macroeconomics Notes* for details.

Problem 1.2

Suppose that the growth rate of some variable, X, is constant and equal to $a > 0$ from time 0 to time t_1; drops to 0 at time t_1; rises gradually from 0 to a from time t_1 to time t_2; and is constant and equal to a after time t_2.

(a) Sketch a graph of the growth rate of X as a function of time.

(b) Sketch a graph of $\ln X$ as a function of time.

Problem 1.3

Describe how, if at all, each of the following developments affects the break-even and actual investment lines in our basic diagram for the Solow model:

(a) The rate of depreciation falls.

(b) The rate of technological progress rises.

(c) The production function is Cobb–Douglas, $f(k) = k^{\alpha}$, and capital's share, α, rises.

(d) Workers exert more effort, so that output per unit of effective labor for a given value of capital per unit of effective labor is higher than before.

Problem 1.4

Consider an economy with technological progress but without population growth that is on its balanced growth path. Now suppose there is a one-time jump in the number of workers.

(a) At the time of the jump, does output per unit of effective labor rise, fall, or stay the same? Why?

(b) After the initial change (if any) in output per unit of effective labor when the new workers appear, is there any further change in output per unit of effective labor? If so, does it rise or fall? Why?

(c) Once the economy has again reached a balanced growth path, is output per unit of effective labor higher, lower, or the same as it was before the new workers appeared? Why?

Problem 1.5

Suppose that the production function is Cobb–Douglas.

(a) Find expressions for k^*, y^*, and c^* as functions of the parameters of the model, s, n, δ, g, and α.

(b) What is the golden-rule value of k?

(c) What saving rate is needed to yield the golden-rule capital stock?

PROBLEM 1.6

Consider a Solow economy that is on its balanced growth path. Assume for simplicity that there is no technological progress. Now suppose that the rate of population growth falls.

(a) What happens to the balanced-growth-path values of capital per worker, output per worker, and consumption per worker? Sketch the paths of these variables as the economy moves to its new balanced growth path.

(b) Describe the effect of the fall in population growth on the path of output (that is, total output, not output per worker).

PROBLEM 1.7

Find the elasticity of output per unit of effective labor on the balanced growth path, y^*, with respect to the rate of population growth, n. If $\alpha_K(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^*?

PROBLEM 1.8

Suppose that investment as a fraction of output in the United States rises permanently from 0.15 to 0.18. Assume that capital's share is $1/3$.

(a) By about how much does output eventually rise relative to what it would have been without the rise in investment?

(b) By about how much does consumption rise relative to what it would have been without the rise in investment?

(c) What is the immediate effect of the rise in investment on consumption? About how long does it take for consumption to return to what it would have been without the rise in investment?

PROBLEM 1.9 *Factor payments in the Solow model.*

Assume that both labor and capital are paid their marginal products. Let w denote $\partial F(K, AL)/\partial L$ and r denote $[\partial F(K, AL)/\partial K] - \delta$.

(a) Show that the marginal product of labor, w, is $A\left[f(k) - kf'(k)\right]$.

(b) Show that if both capital and labor are paid their marginal products, constant returns to scale imply that the total amount paid to the factors of production equals total net output. That is, show that under constant returns, $wL + rK = F(K, AL) - \delta K$.

(c) The return to capital (r) is roughly constant over time, as are the shares of output going to capital and to labor. Does a Solow economy on a balanced growth path exhibit these properties? What are the growth rates of w and r on a balanced growth path?

(d) Suppose the economy begins with a level of k less than k^*. As k moves toward k^*, is w growing at a rate greater than, less than, or equal to its growth rate on the balanced growth path? What about r?

Problem 1.10

(Piketty, 2014; Piketty and Zucman, 2014; Rognlie, 2015) This problem asks you to use a Solow-style model to investigate some ideas that have been discussed in the context of Thomas Piketty's recent work. Consider an economy described by the assumptions of the Solow model, except that factors are paid their marginal products (as in the previous problem), and all labor income is consumed and all other income is saved. Thus, $C(t) = L(t)[\partial Y(t)/\partial L(t)]$.

(a) Show that the properties of the production function and our assumptions about the behavior of L and A imply that the capital-output ratio, K/Y, is rising if and only if the growth rate of K is greater than $n + g$–that is, if and only if k is rising.

(b) Assume that the initial conditions are such that $\partial Y/\partial K$ at $t = 0$ is strictly greater than $n + g + \delta$. Describe the qualitative behavior of the capital-output ratio over time. (For example, does it grow or fall without bound? Gradually approach some constant level from above or below? Something else?) Explain your reasoning.

(c) Many popular summaries of Piketty's work describe his thesis as: Since the return to capital exceeds the growth rate of the economy, the capital-output ratio tends to grow without bound. By the assumptions in part (b), this economy starts in a situation where the return to capital exceeds the economy's growth rate. If you found in (b) that K/Y grows without bound, explain intuitively whether the driving force of this unbounded growth is that the return to capital exceeds the economy's growth rate. Alternatively, if you found in (b) that K/Y does not grow without bound, explain intuitively what is wrong with the statement that the return to capital exceeding the economy's growth rate tends to cause K/Y to grow without bound.

(d) Suppose $F(\cdot)$ is Cobb–Douglas and that the initial situation is as in part (b). Describe the qualitative behavior over time of the share of net capital income (that is, $K(t)[\partial Y(t)/\partial K(t) - \delta]$) in net output (that is, $Y(t) - \delta K(t)$). Explain your reasoning. Is the common statement that an excess of the return to capital over the economy's growth rate causes capital's share to rise over time correct in this case?

Problem 1.11

Consider Problem 1.10. Suppose there is a marginal increase in K.

(a) Derive an expression (in terms of K/Y, δ, the marginal product of capital F_K, and the elasticity of substitution between capital and effective labor in the gross production function $F(\cdot)$) that determines whether a marginal increase in K increases, reduces, or has no effect on the share of net capital income in net output.

(b) Suppose the capital-output ratio is 3, $\delta = 3\%$, and the rate of return on capital $(F_K - \delta) = 5\%$. How large must the elasticity of substitution in the gross production function be for the share of net capital income in net output to rise when K rises?

Problem 1.12

Consider the same setup as at the start of Problem 1.10: the economy is described by the assumptions of the Solow model, except that factors are paid their marginal products and all labor income is consumed and all other income is saved. Show that the economy converges to a balanced growth path, and that the balanced-growth-path level of k equals the golden-rule level of k. What is the intuition for this result?

Problem 1.13

Go through steps analogous to those in equations (1.29)–(1.32) to find how quickly y converges to y^* in the vicinity of the balanced growth path.

Hint: Since $y = f(k)$, we can write $k = g(y)$, where $g(\cdot) = f^{-1}(\cdot)$.

Problem 1.14 *Embodied technological progress.*

(Solow, 1960; Sato, 1966) One view of technological progress is that the productivity of capital goods built at t depends on the state of technology at t and is unaffected by subsequent technological progress. This is known as *embodied technological progress* (technological progress must be "embodied" in new capital before it can raise output). This problem asks you to investigate its effects.

(a) As a preliminary, let us modify the basic Solow model to make technological progress capital-augmenting rather than labor-augmenting. So that a balanced growth path exists, assume that the production function is Cobb–Douglas:

$$Y(t) = [A(t)K(t)]^{\alpha} L(t)^{1-\alpha}$$

Assume that A grows at rate μ: $\dot{A}(t) = \mu A(t)$. Show that the economy converges to a balanced growth path, and find the growth rates of Y and K on the balanced growth path.

Hint: Show that we can write $Y/(A^{\phi}L)$ as a function of $K/(A^{\phi}L)$, where $\phi = \alpha/(1-\alpha)$. Then analyze the dynamics of $K/(A^{\phi}L)$.

(b) Now consider embodied technological progress. Specifically, let the production function be $Y(t) = J(t)^{\alpha} L(t)^{1-\alpha}$, where $J(t)$ is the effective capital stock. The dynamics of $J(t)$ are given by $\dot{J}(t) = sA(t)Y(t) - \delta J(t)$. The presence of the $A(t)$ term in this expression means that the productivity of investment at t depends on the technology at t. Show that the economy converges to a balanced growth path. What are the growth rates of Y and J on the balanced growth path?

Hint: Let $\tilde{J}(t) \equiv J(t)/A(t)$. Then use the same approach as in (a), focusing on $\tilde{J}/(A^{\phi}L)$ instead of $K/(A^{\phi}L)$.

(c) What is the elasticity of output on the balanced growth path with respect to s?

(d) In the vicinity of the balanced growth path, how rapidly does the economy converge to the balanced growth path?

(e) Compare your results for (c) and (d) with the corresponding results in the text for the basic Solow model.

Problem 1.15

Consider a Solow economy on its balanced growth path. Suppose the growth-accounting techniques described in Section 1.7 are applied to this economy.

(a) What fraction of growth in output per worker does growth accounting attribute to growth in capital per worker? What fraction does it attribute to technological progress?

(b) How can you reconcile your results in (a) with the fact that the Solow model implies that the growth rate of output per worker on the balanced growth path is determined solely by the rate of technological progress?

Problem 1.16

(a) In the model of convergence and measurement error in equations (1.39) and (1.40), suppose the true value of b is -1. Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b? Explain.

(b) Suppose there is measurement error in measured 1979 income per capita but not in 1870 income per capita. Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b? Explain.

Problem 1.17

Derive equation (1.50).

Hint: Follow steps analogous to those in equations [1.47] and [1.48].

Chapter 2

Problem 2.1

Consider N firms each with the constant-returns-to-scale production function $Y = F(K, AL)$, or (using the intensive form) $Y = ALf(k)$. Assume $f'(\cdot) > 0$, $f''(\cdot) < 0$. Assume that all firms can hire labor at wage wA and rent capital at cost r, and that all firms have the same value of A.

(a) Consider the problem of a firm trying to produce Y units of output at minimum cost. Show that the cost-minimizing level of k is uniquely defined and is independent of Y, and that all firms therefore choose the same value of k.

(b) Show that the total output of the N cost-minimizing firms equals the output that a single firm with the same production function has if it uses all the labor and capital used by the N firms.

Problem 2.2 *The elasticity of substitution with constant-relative-risk-aversion utility.*

Consider an individual who lives for two periods and whose utility is given by equation (2.43). Let P_1 and P_2 denote the prices of consumption in the two periods, and let W denote the value of the individual's lifetime income; thus the budget constraint is $P_1C_1 + P_2C_2 = W$.

(a) What are the individual's utility-maximizing choices of C_1 and C_2, given P_1, P_2, and W?

(b) The elasticity of substitution between consumption in the two periods is

$$-\frac{P_1/P_2}{C_1/C_2}\frac{\partial(C_1/C_2)}{\partial(P_1/P_2)} \quad \text{or} \quad -\frac{\partial \ln(C_1/C_2)}{\partial \ln(P_1/P_2)}$$

Show that with the utility function (2.43), the elasticity of substitution between C_1 and C_2 is $1/\theta$.

Problem 2.3

(a) Suppose it is known in advance that at some time t_0 the government will confiscate half of whatever wealth each household holds at that time. Does consumption change discontinuously at time t_0? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

(b) Suppose it is known in advance that at t_0 the government will confiscate from each household an amount of wealth equal to half of the wealth of the average household at that time. Does consumption change discontinuously at time t_0? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

Problem 2.4

Assume that the instantaneous utility function $u(C)$ in equation (2.2) is $\ln C$. Consider the problem of a household maximizing (2.2) subject to (2.7). Find an expression for C at each time as a function of initial wealth plus the present value of labor income, the path of $r(t)$, and the parameters of the utility function.

Problem 2.5

Consider a household with utility given by (2.2)–(2.3). Assume that the real interest rate is constant, and let W denote the household's initial wealth plus the present value of its lifetime labor income (the right-hand side of [2.7]). Find the utility-maximizing path of C, given r, W, and the parameters of the utility function.

Problem 2.6 *Growth, saving, and $r - g$.*

Piketty (2014) argues that a fall in the growth rate of the economy is likely to lead to an increase in the difference between the real interest rate and the growth rate. This problem asks you to investigate this issue in the context of the Ramsey–Cass–Koopmans model. Specifically, consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path, and suppose there is a permanent fall in g.

(a) How, if at all, does this affect the $\dot{k} = 0$ curve?

(b) How, if at all, does this affect the $\dot{c} = 0$ curve?

(c) At the time of the change, does c rise, fall, or stay the same, or is it not possible to tell?

(d) At the time of the change, does $r - g$ rise, fall, or stay the same, or is it not possible to tell?

(e) In the long run, does $r - g$ rise, fall, or stay the same, or is it not possible to tell?

(f) Find an expression for the impact of a marginal change in g on the fraction of output that is saved on the balanced growth path. Can one tell whether this expression is positive or negative?

(g) For the case where the production function is Cobb–Douglas, $f(k) = k^{\alpha}$, rewrite your answer to part (f) in terms of ρ, n, g, θ, and α.

Hint: Use the fact that $f'(k^*) = \rho + \theta g$.

Problem 2.7

Describe how each of the following affects the $\dot{c} = 0$ and $\dot{k} = 0$ curves in Figure 2.5, and thus how they affect the balanced-growth-path values of c and k:

(a) A rise in θ.

(b) A downward shift of the production function.

(c) A change in the rate of depreciation from the value of zero assumed in the text to some positive level.

Problem 2.8

Derive an expression analogous to (2.40) for the case of a positive depreciation rate.

Problem 2.9 *A closed-form solution of the Ramsey model.*

(Smith, 2006) Consider the Ramsey model with Cobb–Douglas production, $y(t) = k(t)^{\alpha}$, and with the coefficient of relative risk aversion (θ) and capital's share (α) assumed to be equal.

(a) What is k on the balanced growth path (k^*)?

(b) What is c on the balanced growth path (c^*)?

(c) Let $z(t)$ denote the capital-output ratio,

$$k(t)/y(t)$$

and $x(t)$ denote the consumption-capital ratio,

$$c(t)/k(t)$$

Find expressions for $\dot{z}(t)$ and $\dot{x}(t)/x(t)$ in terms of z, x, and the parameters of the model.

(d) Tentatively conjecture that x is constant along the saddle path. Given this conjecture:

(i) Find the path of z given its initial value, $z(0)$.

(ii) Find the path of y given the initial value of k, $k(0)$. Is the speed of convergence to the balanced growth path, $d \ln\left[y(t) - y^*\right]/dt$, constant as the economy moves along the saddle path?

(e) In the conjectured solution, are the equations of motion for c and k, (2.25) and (2.26), satisfied?

Problem 2.10 *Capital taxation in the Ramsey–Cass–Koopmans model.*

Consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path. Suppose that at some time, which we will call time 0, the government switches to a policy of taxing investment income at rate τ. Thus the real interest rate that households face is now given by $r(t) = (1-\tau) f'(k(t))$. Assume that the government returns the revenue it collects from this tax through lump-sum transfers. Finally, assume that this change in tax policy is unanticipated.

(a) How, if at all, does the tax affect the $\dot{c} = 0$ locus? The $\dot{k} = 0$ locus?

(b) How does the economy respond to the adoption of the tax at time 0? What are the dynamics after time 0?

(c) How do the values of c and k on the new balanced growth path compare with their values on the old balanced growth path?

(d) (Barro, Mankiw, and Sala-i-Martin, 1995) Suppose there are many economies like this one. Workers' preferences are the same in each country, but the tax rates on investment income may vary across countries. Assume that each country is on its balanced growth path.

(i) Show that the saving rate on the balanced growth path, $\left(y^* - c^*\right)/y^*$, is decreasing in τ.

(ii) Do citizens in low-τ, high-k^*, high-saving countries have any incentive to invest in low-saving countries? Why or why not?

(e) Does your answer to part (c) imply that a policy of *subsidizing investment* (that is, making $\tau < 0$), and raising the revenue for this subsidy through lump-sum taxes, increases welfare? Why or why not?

(f) How, if at all, do the answers to parts (a) and (b) change if the government does not rebate the revenue from the tax but instead uses it to make government purchases?

PROBLEM 2.11 *Using the phase diagram to analyze the impact of an anticipated change.*

Consider the policy described in Problem 2.10, but suppose that instead of announcing and implementing the tax at time 0, the government announces at time 0 that at some later time, time t_1, investment income will begin to be taxed at rate τ.

(a) Draw the phase diagram showing the dynamics of c and k after time t_1.

(b) Can c change discontinuously at time t_1? Why or why not?

(c) Draw the phase diagram showing the dynamics of c and k before t_1.

(d) In light of your answers to parts (a), (b), and (c), what must c do at time 0?

(e) Summarize your results by sketching the paths of c and k as functions of time.

PROBLEM 2.12 *Using the phase diagram to analyze the impact of unanticipated and anticipated temporary changes.*

Analyze the following two variations on Problem 2.11:

(a) At time 0, the government announces that it will tax investment income at rate τ from time 0 until some later date t_1; thereafter investment income will again be untaxed.

(b) At time 0, the government announces that from time t_1 to some later time t_2, it will tax investment income at rate τ; before t_1 and after t_2, investment income will not be taxed.

PROBLEM 2.13 *An interesting situation in the Ramsey–Cass–Koopmans model.*

(a) Consider the Ramsey–Cass–Koopmans model where k at time 0 (which–as always–the model takes as given) is at the golden-rule level: $k(0) = k^{GR}$. Sketch the paths of c and k.

(b) Consider the same initial situation as in part (a), but in the version of the model that includes government purchases. Assume that G is constant and equal $\bar{G}$. Crucially, $\bar{G}$ is strictly less than $f\left(k^{GR}\right) - (n+g)\,k^{GR}$ and strictly greater than $f(k^*) - (n+g)\,k^*$ (where k^* is the level of k on the balanced growth path the economy would have if G were constant and equal to 0). Sketch the paths of c and k.

Hint: This problem is hard but can be answered by thinking things through slowly and carefully.

Problem 2.14

Consider the Diamond model with logarithmic utility and Cobb–Douglas production. Describe how each of the following affects k_{t+1} as a function of k_t:

(a) A rise in n.

(b) A downward shift of the production function (that is, $f(k)$ takes the form $B k^{\alpha}$, and B falls).

(c) A rise in α.

Problem 2.15 *A discrete-time version of the Solow model.*

Suppose $Y_t = F(K_t, A_t L_t)$, with $F(\cdot)$ having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g) A_t$, $L_{t+1} = (1+n) L_t$, and $K_{t+1} = K_t + s Y_t - \delta K_t$.

(a) Find an expression for k_{t+1} as a function of k_t.

(b) Sketch k_{t+1} as a function of k_t. Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path?

(c) Find an expression for consumption per unit of effective labor on the balanced growth path as a function of the balanced-growth-path value of k. What is the marginal product of capital, $f'(k)$, when k maximizes consumption per unit of effective labor on the balanced growth path?

(d) Assume that the production function is Cobb–Douglas.

(i) What is k_{t+1} as a function of k_t?

(ii) What is k^*, the value of k on the balanced growth path?

(iii) Along the lines of equations (2.65)–(2.67), in the text, linearize the expression in subpart (i) around $k_t = k^*$, and find the rate of convergence of k to k^*.

Problem 2.16 *Depreciation in the Diamond model and microeconomic foundations for the Solow model.*

Suppose that in the Diamond model capital depreciates at rate δ, so that $r_t = f'(k_t) - \delta$.

(a) How, if at all, does this change in the model affect equation (2.60) giving k_{t+1} as a function of k_t?

(b) In the special case of logarithmic utility, Cobb–Douglas production, and $\delta = 1$, what is the equation for k_{t+1} as a function of k_t? Compare this with the analogous expression for the discrete-time version of the Solow model with $\delta = 1$ from part (a) of Problem 2.15.

Problem 2.17 *Social security in the Diamond model.*

Consider a Diamond economy where g is zero, production is Cobb–Douglas, and utility is logarithmic.

(a) **Pay-as-you-go social security.** Suppose the government taxes each young individual an amount T and uses the proceeds to pay benefits to old individuals; thus each old person receives $(1+n)T$.

(i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t?

(ii) How, if at all, does this change affect the balanced-growth-path value of k?

(iii) If the economy is initially on a balanced growth path that is dynamically efficient, how does a marginal increase in T affect the welfare of current and future generations? What happens if the initial balanced growth path is dynamically inefficient?

(b) **Fully funded social security.** Suppose the government taxes each young person an amount T and uses the proceeds to purchase capital. Individuals born at t therefore receive $(1+r_{t+1})T$ when they are old.

(i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t?

(ii) How, if at all, does this change affect the balanced-growth-path value of k?

Problem 2.18 *The basic overlapping-generations model.*

(Samuelson, 1958; Allais, 1947) Suppose, as in the Diamond model, that L_t two-period-lived individuals are born in period t and that $L_t = (1+n)L_{t-1}$. For simplicity, let utility be logarithmic with no discounting: $U_t = \ln(C_{1t}) + \ln(C_{2t+1})$.

The production side of the economy is simpler than in the Diamond model. Each individual born at time t is endowed with A units of the economy's single good. The good can be either consumed or stored. Each unit stored yields $x > 0$ units of the good in the following period.[1]

Finally, assume that in the initial period, period 0, in addition to the L_0 young indi-

[1] Note that this is the same as the Diamond economy with $g = 0$, $F(K_t, AL_t) = AL_t + xK_t$, and $\delta = 1$. With this production function, since individuals supply 1 unit of labor when they are young, an individual born in t obtains A units of the good. And each unit saved yields $1+r = 1+\partial F(K, AL)/\partial K - \delta = 1 + x - 1 = x$ units of second-period consumption.

viduals each endowed with A units of the good, there are $[1/(1+n)]L_0$ individuals who are alive only in period 0. Each of these "old" individuals is endowed with some amount Z of the good; their utility is simply their consumption in the initial period, C_{20}.

(a) Describe the decentralized equilibrium of this economy.

Hint: Given the overlapping-generations structure, will the members of any generation engage in transactions with members of another generation?

(b) Consider paths where the fraction of agents' endowments that is stored, f_t, is constant over time. What is total consumption (that is, consumption of all the young plus consumption of all the old) per person on such a path as a function of f? If $x < 1+n$, what value of f satisfying $0 \leq f \leq 1$ maximizes consumption per person? Is the decentralized equilibrium Pareto efficient in this case? If not, how can a social planner raise welfare?

Problem 2.19 *Stationary monetary equilibria in the Samuelson OLG model.*

(Samuelson, 1958) Consider the setup described in Problem 2.18. Assume that $x < 1+n$. Suppose that the old individuals in period 0, in addition to being endowed with Z units of the good, are each endowed with M units of a storable, divisible commodity, which we will call money. Money is not a source of utility.

(a) Consider an individual born at t. Suppose the price of the good in units of money is P_t in t and P_{t+1} in $t+1$. Thus the individual can sell units of endowment for P_t units of money and then use that money to buy P_t/P_{t+1} units of the next generation's endowment the following period. What is the individual's behavior as a function of P_t/P_{t+1}?

(b) Show that there is an equilibrium with $P_{t+1} = P_t/(1+n)$ for all $t \geq 0$ and no storage, and thus that the presence of "money" allows the economy to reach the golden-rule level of storage.

(c) Show that there are also equilibria with $P_{t+1} = P_t/x$ for all $t \geq 0$.

(d) Finally, explain why $P_t = \infty$ for all t (that is, money is worthless) is also an equilibrium. Explain why this is the only equilibrium if the economy ends at some date, as in Problem 2.20(b) below.

Hint: Reason backward from the last period.

Problem 2.20 *The source of dynamic inefficiency.*

(Shell, 1971) There are two ways in which the Diamond and Samuelson models differ from textbook models. First, markets are incomplete: because individuals cannot trade with individuals who have not been born, some possible transactions are ruled out. Second, because time goes on forever, there are an infinite number of agents. This problem asks you to investigate which of these is the source of the possibility of dynamic inefficiency. For simplicity, it focuses on the Samuelson overlapping-generations model (see the previous two problems), again with log utility and no discounting. To simplify further, it assumes $n = 0$ and $0 < x < 1$.

(a) **Incomplete markets.** Suppose we eliminate incomplete markets from the model by allowing all agents to trade in a competitive market "before" the beginning of time. That is, a Walrasian auctioneer calls out prices $Q_0, Q_1, Q_2, \ldots$ for the good at each date. Individuals can then make sales and purchases at these prices given their endowments and their ability to store. The budget constraint of an individual born at t is thus $Q_t C_{1t} + Q_{t+1} C_{2t+1} = Q_t (A - S_t) + Q_{t+1} x S_t$, where S_t (which must satisfy $0 \leq S_t \leq A$) is the amount the individual stores.

(i) Suppose the auctioneer announces $Q_{t+1} = Q_t/x$ for all $t > 0$. Show that in this case individuals are indifferent concerning how much to store, that there is a set of storage decisions such that markets clear at every date, and that this equilibrium is the same as the equilibrium described in part (a) of Problem 2.18.

(ii) Suppose the auctioneer announces prices that fail to satisfy $Q_{t+1} = Q_t/x$ at some date. Show that at the first date that does not satisfy this condition the market for the good cannot clear, and thus that the proposed price path cannot be an equilibrium.

(b) **Infinite duration.** Suppose that the economy ends at some date T. That is, suppose the individuals born at T live only one period (and hence seek to maximize C_{1T}), and that thereafter no individuals are born. Show that the decentralized equilibrium is Pareto efficient.

(c) In light of these answers, is it incomplete markets or infinite duration that is the source of dynamic inefficiency?

Problem 2.21 *Explosive paths in the Samuelson overlapping-generations model.*

(Black, 1974; Brock, 1975; Calvo, 1978) Consider the setup described in Problem 2.19. Assume that x is zero, and assume that utility is constant-relative-risk-aversion with $\theta < 1$ rather than logarithmic. Finally, assume for simplicity that $n = 0$.

(a) What is the behavior of an individual born at t as a function of P_t/P_{t+1}? Show that the amount of his or her endowment that the individual sells for money is an increasing function of P_t/P_{t+1} and approaches zero as this ratio approaches zero.

(b) Suppose $P_0/P_1 < 1$. How much of the good are the individuals born in period 0 planning to buy in period 1 from the individuals born then? What must P_1/P_2 be for the individuals born in period 1 to want to supply this amount?

(c) Iterating this reasoning forward, what is the qualitative behavior of P_t/P_{t+1} over time? Does this represent an equilibrium path for the economy?

(d) Can there be an equilibrium path with $P_0/P_1 > 1$?

Chapter 3

Problem 3.1

Consider the model of Section 3.2 with $\theta < 1$.

(a) On the balanced growth path, $\dot{A} = g_A^* A(t)$, where g_A^* is the balanced-growth-path value of g_A. Use this fact and equation (3.6) to derive an expression for $A(t)$ on the balanced growth path in terms of B, a_L, γ, θ, and $L(t)$.

(b) Use your answer to part (a) and the production function, (3.5), to obtain an expression for $Y(t)$ on the balanced growth path. Find the value of a_L that maximizes output on the balanced growth path.

Problem 3.2

Consider two economies (indexed by $i = 1, 2$) described by $Y_i(t) = K_i(t)^\theta$ and $\dot{K}_i(t) = s_i Y_i(t)$, where $\theta > 1$. Suppose that the two economies have the same initial value of K, but that $s_1 > s_2$. Show that Y_1/Y_2 is continually rising.

Problem 3.3

Consider the economy analyzed in Section 3.3. Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:

(a) An increase in n.

(b) An increase in a_K.

(c) An increase in θ.

Problem 3.4

Consider the economy described in Section 3.3, and assume $\beta + \theta < 1$ and $n > 0$. Suppose the economy is initially on its balanced growth path, and that there is a permanent increase in s.

(a) How, if at all, does the change affect the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines? How, if at all, does it affect the location of the economy in (g_A, g_K) space at the time of the change?

(b) What are the dynamics of g_A and g_K after the increase in s? Sketch the path of log output per worker.

(c) Intuitively, how does the effect of the increase in s compare with its effect in the Solow model?

Problem 3.5

Consider the model of Section 3.3 with $\beta + \theta = 1$ and $n = 0$.

(a) Using (3.14) and (3.16), find the value that A/K must have for g_K and g_A to be equal.

(b) Using your result in part (a), find the growth rate of A and K when $g_K = g_A$.

(c) How does an increase in s affect the long-run growth rate of the economy?

(d) What value of a_K maximizes the long-run growth rate of the economy? Intuitively, why is this value not increasing in β, the importance of capital in the R&D sector?

Problem 3.6

Consider the model of Section 3.3 with $\beta + \theta > 1$ and $n > 0$.

(a) Draw the phase diagram for this case.

(b) Show that regardless of the economy's initial conditions, eventually the growth rates of A and K (and hence the growth rate of Y) are increasing continually.

(c) Repeat parts (a) and (b) for the case of $\beta + \theta = 1, n > 0$.

Problem 3.7 *Learning-by-doing.*

Suppose that output is given by equation (3.22), $Y(t) = K(t)^{\alpha}[A(t)L(t)]^{1-\alpha}$; that L is constant and equal to 1; that $\dot{K}(t) = sY(t)$; and that knowledge accumulation occurs as a side effect of goods production: $\dot{A}(t) = BY(t)$.

(a) Find expressions for $g_A(t)$ and $g_K(t)$ in terms of $A(t)$, $K(t)$, and the parameters.

(b) Sketch the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines in (g_A, g_K) space.

(c) Does the economy converge to a balanced growth path? If so, what are the growth rates of K, A, and Y on the balanced growth path?

(d) How does an increase in s affect long-run growth?

Problem 3.8

Consider the model of Section 3.5. Suppose, however, that households have constant-relative-risk-aversion utility with a coefficient of relative risk aversion of θ. Find the equilibrium level of labor in the R&D sector, L_A.

Problem 3.9

Suppose that policymakers, realizing that monopoly power creates distortions, put controls on the prices that patent-holders in the Romer model can charge for the inputs embodying their ideas. Specifically, suppose they require patent-holders to charge $\delta w(t)/\phi$, where δ satisfies $\phi \leq \delta \leq 1$.

(a) What is the equilibrium growth rate of the economy as a function of δ and the other parameters of the model? Does a reduction in δ increase, decrease, or have no effect on the equilibrium growth rate, or is it not possible to tell?

(b) Explain intuitively why setting $\delta = \phi$, thereby requiring patent-holders to charge marginal cost and so eliminating the monopoly distortion, does not maximize social welfare.

Problem 3.10

(a) Show that (3.48) follows from (3.47).

(b) Derive (3.49).

Problem 3.11 *The balanced growth path of a semi-endogenous version of the Romer model.*

(Jones, 1995) Consider the model of Section 3.5 with two changes. First, existing knowledge contributes less than proportionally to the production of new knowledge, as in Case 1 of the model of Section 3.2: $\dot{A}(t) = BL_A(t)A(t)^{\theta}$, $\theta < 1$. Second, population is growing at rate n rather than constant: $L(t) = L(0)e^{nt}$, $n > 0$. (Consistent with this, assume that utility is given by equation [2.2] with $u(\cdot)$ logarithmic.) The analysis of Section 3.2 implies that such a model will exhibit transition dynamics rather than always immediately being on a balanced growth path. This problem therefore asks you not to analyze the full dynamics of the model, but to focus on its properties when it is on a balanced growth path. Specifically, it looks at situations where the fraction of the labor force engaged in R&D, a_L, is constant, and all variables of the model are growing at constant rates.

Hint: You are welcome to assume rather than derive that on a balanced growth path, the wage and consumption per person grow at the same rate as Y/L.

(a) What are balanced-growth-path values of the growth rates of Y and Y/L and of

r as functions of the balanced-growth-path value of the growth rate of A and parameters of the model?

(b) On the balanced growth path, what is the present value of the profits from the discovery of a new idea at time t as a function of $L(t)$, $A(t)$, $w(t)$, and exogenous parameters?

(c) What is $\dot{A}(t)/A(t)$ on the balanced growth path as a function of a_L and exogenous parameters? What is $L(t)A(t)^{\theta-1}$ on the balanced growth path?

Hint: Consider equations [3.6] and [3.7] in the case of $\gamma = 1$.

(d) What is a_L on the balanced growth path?

(e) Discuss how changes in each of ρ, B, ϕ, n, and θ affect the balanced-growth-path value of a_L. In the cases of ρ, B, and ϕ, are the effects in the same direction as the effects on L_A in the model of Section 3.5?

Problem 3.12 *Learning-by-doing with microeconomic foundations.*

Consider a variant of the model in equations (3.22)–(3.25). Suppose firm i's output is $Y_i(t) = K_i(t)^{\alpha}[A(t)L_i(t)]^{1-\alpha}$, and that $A(t) = BK(t)$. Here K_i and L_i are the amounts of capital and labor used by firm i, and K is the aggregate capital stock. Capital and labor earn their *private* marginal products. As in the model of Section 3.5, the economy is populated by infinitely lived households that own the economy's initial capital stock. The utility of the representative household takes the constant-relative-risk-aversion form in equations (2.2)–(2.3). Population growth is zero.

(a) (i) What are the private marginal products of capital and labor at firm i as functions of $K_i(t)$, $L_i(t)$, $K(t)$, and the parameters of the model?

(ii) Explain why the capital-labor ratio must be the same at all firms, so

$$\frac{K_i(t)}{L_i(t)} = \frac{K(t)}{L(t)}$$

(iii) What are $w(t)$ and $r(t)$ as functions of $K(t)$, L, and the parameters of the model?

(b) What must the growth rate of consumption be in equilibrium? Assume for simplicity that the parameter values are such that the growth rate is strictly positive and less than the interest rate. Sketch an explanation of why the equilibrium growth rate of output equals the equilibrium growth rate of consumption.

Hint: Consider equation [2.22].

(c) Describe how long-run growth is affected by:

(i) A rise in B.

(ii) A rise in ρ.

(iii) A rise in L.

(d) Is the equilibrium growth rate more than, less than, or equal to the socially optimal rate, or is it not possible to tell?

Problem 3.13

(Rebelo, 1991) Assume that there are two sectors, one producing consumption goods and one producing capital goods, and two factors of production: capital and land. Capital is used in both sectors, but land is used only in producing consumption goods. Specifically, the production functions are $C(t) = K_C(t)^{\alpha} T^{1-\alpha}$ and $\dot{K}(t) = BK_K(t)$, where K_C and K_K are the amounts of capital used in the two sectors (so $K_C(t)+K_K(t) = K(t)$) and T is the amount of land, and $0 < \alpha < 1$ and $B > 0$. Factors are paid their marginal products, and capital can move freely between the two sectors. T is normalized to 1 for simplicity.

(a) Let $P_K(t)$ denote the price of capital goods relative to consumption goods at time t. Use the fact that the earnings of capital in units of consumption goods in the two sectors must be equal to derive a condition relating $P_K(t)$, $K_C(t)$, and the parameters α and B. If K_C is growing at rate $g_K(t)$, at what rate must P_K be growing (or falling)? Let $g_P(t)$ denote this growth rate.

(b) The real interest rate in terms of consumption is $B + g_P(t)$.[2] Thus, assuming that households have our standard utility function, (2.22)–(2.23), the growth rate of consumption must be $\left(B + g_P - \rho\right)/\theta \equiv g_C$. Assume $\rho < B$.

(i) Use your results in part (a) to express $g_C(t)$ in terms of $g_K(t)$ rather than $g_P(t)$.

(ii) Given the production function for consumption goods, at what rate must K_C be growing for C to be growing at rate $g_C(t)$?

(iii) Combine your answers to (i) and (ii) to solve for $g_K(t)$ and $g_C(t)$ in terms of the underlying parameters.

(c) Suppose that investment income is taxed at rate τ, so that the real interest rate households face is $(1-\tau)\left(B + g_P\right)$. How, if at all, does τ affect the equilibrium growth rate of consumption?

[2] To see this, note that capital in the investment sector produces new capital at rate B and changes in value relative to the consumption good at rate g_P. (Because the return to capital is the same in the two sectors, the same must be true of capital in the consumption sector.)

Problem 3.14 *Delays in the transmission of knowledge to poor countries.*

(a) Assume that the world consists of two regions, the North and the South. The North is described by $Y_N(t) = A_N(t)(1-a_L)L_N$ and $\dot{A}_N(t) = a_L L_N A_N(t)$. The South does not do R&D but simply uses the technology developed in the North; however, the technology used in the South lags the North's by τ years. Thus $Y_S(t) = A_S(t)L_S$ and $A_S(t) = A_N(t-\tau)$. If the growth rate of output per worker in the North is 3 percent per year, and if a_L is close to 0, what must τ be for output per worker in the North to exceed that in the South by a factor of 10?

(b) Suppose instead that both the North and the South are described by the Solow model: $y_i(t) = f(k_i(t))$, where

$$y_i(t) \equiv \frac{Y_i(t)}{L_i(t)} \quad \text{and} \quad k_i(t) \equiv \frac{K_i(t)}{L_i(t)}, \qquad i = N, S$$

As in the Solow model, assume $\dot{K}_i(t) = sY_i(t) - \delta K_i(t)$ and $\dot{L}_i(t) = nL_i(t)$; the two countries are assumed to have the same saving rates and rates of population growth. Finally, $\dot{A}_N(t) = gA_N(t)$ and $A_S(t) = A_N(t-\tau)$.

(i) Show that the value of k on the balanced growth path, k^*, is the same for the two countries.

(ii) Does introducing capital change the answer to part (a)? Explain. (Continue to assume $g = 3\%$.)

Problem 3.15

Which of the following possible regression results concerning the elasticity of long-run output with respect to the saving rate would provide the best evidence that differences in saving rates are not important to cross-country income differences? (1) A point estimate of 5 with a standard error of 2; (2) a point estimate of 0.1 with a standard error of 0.01; (3) a point estimate of 0.001 with a standard error of 5; (4) a point estimate of -2 with a standard error of 5. Explain your answer.

Hint: See the discussion of confidence intervals versus t-statistics in Section 3.6.

Chapter 4

Problem 4.1 *The golden-rule level of education.*

Consider the model of Section 4.1 with the assumption that $G(E)$ takes the form $G(E) = e^{\phi E}$.

(a) Find an expression that characterizes the value of E that maximizes the level of output per person on the balanced growth path. Are there cases where this value equals 0? Are there cases where it equals T?

(b) Assuming an interior solution, describe how, if at all, the golden-rule level of E (that is, the level of E you characterized in part (a)) is affected by each of the following changes:

(i) A rise in T.

(ii) A fall in n.

Problem 4.2 *Endogenizing the choice of E.*

(Bils and Klenow, 2000) Suppose that the wage of a worker with education E at time t is $be^{gt}e^{\phi E}$. Consider a worker born at time 0 who will be in school for the first E years of life and will work for the remaining $T - E$ years. Assume that the interest rate is constant and equal to $\bar{r}$.

(a) What is the present discounted value of the worker's lifetime earnings as a function of E, T, b, $\bar{r}$, ϕ, and g?

(b) Find the first-order condition for the value of E that maximizes the expression you found in part (a). Let E^* denote this value of E. (Assume an interior solution.)

(c) Describe how each of the following developments affects E^*:

(i) A rise in T.

(ii) A rise in $\bar{r}$.

(iii) A rise in g.

Problem 4.3

Suppose output in country i is given by $Y_i = A_i Q_i e^{\phi E_i} L_i$. Here E_i is each worker's years of education, Q_i is the quality of education, and the rest of the notation is standard. Higher output per worker raises the quality of education. Specifically, Q_i is given by $B_i (Y_i/L_i)^{\gamma}$, $0 < \gamma < 1$, $B_i > 0$.

Our goal is to decompose the difference in log output per worker between two countries, 1 and 2, into the contributions of education and all other forces. We have data on Y, L, and E in the two countries, and we know the values of the parameters ϕ and γ.

(a) Explain in what way attributing amount $\phi(E_2 - E_1)$ of $\ln(Y_2/L_2) - \ln(Y_1/L_1)$ to education and the remainder to other forces would understate the contribution of education to the difference in log output per worker between the two countries.

(b) What would be a better measure of the contribution of education to the difference in log output per worker?

Problem 4.4

Suppose the production function is $Y = K^{\alpha}\left(e^{\phi E} L\right)^{1-\alpha}$, $0 < \alpha < 1$. E is the amount of education workers receive; the rest of the notation is standard. Assume that there is perfect capital mobility. In particular, K always adjusts so that the marginal product of capital equals the world rate of return, r^*.

(a) Find an expression for the marginal product of capital as a function of K, E, L, and the parameters of the production function.

(b) Use the equation you derived in part (a) to find K as a function of r^*, E, L, and the parameters of the production function.

(c) Use your answer in part (b) to find an expression for $d(\ln Y)/dE$, incorporating the effect of E on Y via K.

(d) Explain intuitively how capital mobility affects the impact of the change in E on output.

Problem 4.5

(Mankiw, Romer, and Weil, 1992) Suppose output is given by

$$Y(t) = K(t)^{\alpha} H(t)^{\beta} [A(t) L(t)]^{1-\alpha-\beta}, \qquad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < 1.$$

Here L is the number of workers and H is their total amount of skills. The remainder of the notation is standard. The dynamics of the inputs are $\dot{L}(t) = nL(t)$, $\dot{A}(t) = gA(t)$, $\dot{K}(t) = s_K Y(t) - \delta K(t)$, $\dot{H}(t) = s_H Y(t) - \delta H(t)$, where $0 < s_K < 1$, $0 < s_H < 1$, and

$n+g+\delta>0$. $L(0)$, $A(0)$, $K(0)$, and $H(0)$ are given, and are all strictly positive. Finally, define $y(t) \equiv Y(t)/[A(t)L(t)]$, $k(t) \equiv K(t)/[A(t)L(t)]$, and $h(t) \equiv H(t)/[A(t)L(t)]$.

(a) Derive an expression for $y(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model.

(b) Derive an expression for $\dot{k}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k,h) space, sketch the set of points where $\dot{k}=0$.

(c) Derive an expression for $\dot{h}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k,h) space, sketch the set of points where $\dot{h}=0$.

(d) Does the economy converge to a balanced growth path? Why or why not? If so, what is the growth rate of output per worker on the balanced growth path? If not, in general terms what is the behavior of output per worker over time?

Problem 4.6

Consider the model in Problem 4.5.

(a) What are the balanced-growth-path values of k and h in terms of s_K, s_H, and the other parameters of the model?

(b) Suppose $\alpha = 1/3$ and $\beta = 1/2$. Consider two countries, A and B, and suppose that both s_K and s_H are twice as large in Country A as in Country B and that the countries are otherwise identical. What is the ratio of the balanced-growth-path value of income per worker in Country A to its value in Country B implied by the model?

(c) Consider the same assumptions as in part (b). What is the ratio of the balanced-growth-path value of skills per worker in Country A to its value in Country B implied by the model?

Problem 4.7

(Jones, 2002) Consider the model of Section 4.1 with the assumption that $G(E)=e^{\phi E}$. Suppose, however, that E, rather than being constant, is increasing steadily: $\dot{E}(t)=m$, where $m>0$. Assume that, despite the steady increase in the amount of education people are getting, the growth rate of the number of workers is constant and equal to n, as in the basic model.

(a) With this change in the model, what is the long-run growth rate of output per worker?

(b) In the United States over the past century, if we measure E as years of schooling, $\phi \simeq 0.1$ and $m \simeq 1/15$. Overall growth of output per worker has been about 2 percent per year. In light of your answer to (a), approximately what fraction of this overall growth has been due to increasing education?

(c) Can $\dot{E}(t)$ continue to equal $m > 0$ forever? Explain.

Problem 4.8

Consider the following model with physical and human capital:

$$
\begin{aligned}
Y(t) &= [(1 - a_K) K(t)]^{\alpha} [(1 - a_H) H(t)]^{1-\alpha} && 0 < \alpha, a_K, a_H < 1 \\
\dot{K}(t) &= sY(t) - \delta_K K(t) \\
\dot{H}(t) &= B [a_K K(t)]^{\gamma} [a_H H(t)]^{\phi} [A(t) L(t)]^{1-\gamma-\phi} - \delta_H H(t) && \gamma, \phi > 0,\ \gamma + \phi < 1 \\
\dot{L}(t) &= nL(t) \\
\dot{A}(t) &= gA(t),
\end{aligned}
$$

where a_K and a_H are the fractions of the stocks of physical and human capital used in the education sector.

This model assumes that human capital is produced in its own sector with its own production function. Bodies (L) are useful only as something to be educated, not as an input into the production of final goods. Similarly, knowledge (A) is useful only as something that can be conveyed to students, not as a direct input to goods production.

(a) Define $k \equiv K/(AL)$ and $h \equiv H/(AL)$. Derive equations for $\dot{k}$ and $\dot{h}$.

(b) Find an equation describing the set of combinations of h and k such that $\dot{k} = 0$. Sketch in (h, k) space. Do the same for $\dot{h} = 0$.

(c) Does this economy have a balanced growth path? If so, is it unique? Is it stable? What are the growth rates of output per person, physical capital per person, and human capital per person on the balanced growth path?

(d) Suppose the economy is initially on a balanced growth path, and that there is a permanent increase in s. How does this change affect the path of output per person over time?

Problem 4.9 *Increasing returns in a model with human capital.*

(Lucas, 1988) Suppose that $Y(t) = K(t)^{\alpha} [(1 - a_H) H(t)]^{\beta}$, $\dot{H}(t) = Ba_H H(t)$, and $\dot{K}(t) = sY(t)$. Assume $0 < \alpha < 1$, $0 < \beta < 1$, and $\alpha + \beta > 1$.

(a) What is the growth rate of H?

(b) Does the economy converge to a balanced growth path? If so, what are the growth rates of K and Y on the balanced growth path?

Problem 4.10 *A different form of measurement error.*

Suppose the true relationship between social infrastructure (SI) and log income per person (y) is $y_i = a + bSI_i + e_i$. There are two components of social infrastructure, SI_i^A and SI_i^B (with $SI_i = SI_i^A + SI_i^B$), and we only have data on one of the components, SI_i^A. Both SI_i^A and SI_i^B are uncorrelated with e. We are considering running an OLS regression of y on a constant and SI^A.

(a) Derive an expression of the form, $y_i = a + bSI_i^A + \text{other terms}$.

(b) Use your answer to part (a) to determine whether an OLS regression of y on a constant and SI^A will produce an unbiased estimate of the impact of social infrastructure on income if:

  (i) SI^A and SI^B are uncorrelated.

  (ii) SI^A and SI^B are positively correlated.

Problem 4.11

Briefly explain whether each of the following statements concerning a cross-country regression of income per person on a measure of social infrastructure is true or false:

(a) "If the regression is estimated by ordinary least squares, it shows the effect of social infrastructure on output per person."

(b) "If the regression is estimated by instrumental variables using variables that are not affected by social infrastructure as instruments, it shows the effect of social infrastructure on output per person."

(c) "If the regression is estimated by ordinary least squares and has a high R^2, this means that there are no important influences on output per person that are omitted from the regression; thus in this case, the coefficient estimate from the regression is likely to be close to the true effect of social infrastructure on output per person."

Problem 4.12 *Convergence regressions.*

(a) **Convergence.** Let y_i denote log output per worker in country i. Suppose all countries have the same balanced-growth-path level of log income per worker, y^*. Suppose also that y_i evolves according to $\mathrm{d}y_i(t)/dt = -\lambda\left[y_i(t) - y^*\right]$.

  (i) What is $y_i(t)$ as a function of $y_i(0)$, y^*, λ, and t?

  (ii) Suppose that $y_i(t)$ in fact equals the expression you derived in part (i) plus a mean-zero random disturbance that is uncorrelated with $y_i(0)$. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$.

What is the relation between β, the coefficient on $y_i(0)$ in the regression, and λ, the speed of convergence? Given this, how could you estimate λ from an estimate of β?

Hint: For a univariate OLS regression, the coefficient on the right-hand-side variable equals the covariance between the right-hand-side and left-hand-side variables divided by the variance of the right-hand-side variable.

(iii) If β in part (ii) is negative (so that rich countries on average grow less than poor countries), is $\mathrm{Var}\left(y_i(t)\right)$ necessarily less than $\mathrm{Var}\left(y_i(0)\right)$, so that the cross-country variance of income is falling? Explain. If β is positive, is $\mathrm{Var}\left(y_i(t)\right)$ necessarily more than $\mathrm{Var}\left(y_i(0)\right)$? Explain.

(b) **Conditional convergence.** Suppose $y_i^* = a + bX_i$, and that

$$\frac{\mathrm{d}y_i(t)}{\mathrm{d}t} = -\lambda\left[y_i(t) - y_i^*\right]$$

(i) What is $y_i(t)$ as a function of $y_i(0)$, X_i, λ, and t?

(ii) Suppose that $y_i(0) = y_i^* + u_i$ and that $y_i(t)$ equals the expression you derived in part (i) plus a mean-zero random disturbance, e_i, where X_i, u_i, and e_i are uncorrelated with one another. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$. Suppose one attempts to infer λ from the estimate of β using the formula in part (a)(ii). Will this lead to a correct estimate of λ, an overestimate, or an underestimate?

(iii) Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \gamma X_i + \varepsilon_i$. Under the same assumptions as in part (ii), how could one estimate b, the effect of X on the balanced-growth-path value of y, from estimates of β and γ?

Chapter 5

Problem 5.1

Redo the calculations reported in Table 5.1, 5.2, or 5.3 for any country other than the United States.

Problem 5.2

Redo the calculations reported in Table 5.3 for the following:

(a) Employees' compensation as a share of national income.

(b) The labor force participation rate.

(c) The federal government budget deficit as a share of GDP.

(d) The Standard and Poor's 500 composite stock price index.

(e) The difference in yields between Moody's Baa and Aaa bonds.

(f) The difference in yields between 10-year and 3-month U.S. Treasury securities.

(g) The weighted average exchange rate of the U.S. dollar against major currencies.

Problem 5.3

Let A_0 denote the value of A in period 0, and let the behavior of $\ln A$ be given by equations (5.8)–(5.9).

(a) Express $\ln A_1$, $\ln A_2$, and $\ln A_3$ in terms of $\ln A_0$, ε_{A1}, ε_{A2}, ε_{A3}, $\bar{A}$, and g.

(b) In light of the fact that the expectations of the ε_A's are zero, what are the expectations of $\ln A_1$, $\ln A_2$, and $\ln A_3$ given $\ln A_0$, $\bar{A}$, and g?

Problem 5.4

Suppose the period-t utility function, u_t, is $u_t = \ln c_t + b(1-\ell_t)^{1-\gamma}/(1-\gamma)$, $b > 0$, $\gamma > 0$, rather than (5.7).

(a) Consider the one-period problem analogous to that investigated in (5.12)–(5.15). How, if at all, does labor supply depend on the wage?

(b) Consider the two-period problem analogous to that investigated in (5.16)–(5.21). How does the relative demand for leisure in the two periods depend on the relative wage? How does it depend on the interest rate? Explain intuitively why γ affects the responsiveness of labor supply to wages and the interest rate.

Problem 5.5

Consider the problem investigated in (5.16)–(5.21).

(a) Show that an increase in both w_1 and w_2 that leaves w_1/w_2 unchanged does not affect ℓ_1 or ℓ_2.

(b) Now assume that the household has initial wealth of amount $Z > 0$.

(i) Does (5.23) continue to hold? Why or why not?

(ii) Does the result in (a) continue to hold? Why or why not?

Problem 5.6

Suppose an individual lives for two periods and has utility $\ln C_1 + \ln C_2$.

(a) Suppose the individual has labor income of Y_1 in the first period of life and zero in the second period. Second-period consumption is thus $(1+r)(Y_1 - C_1)$; r, the rate of return, is potentially random.

(i) Find the first-order condition for the individual's choice of C_1.

(ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?

(b) Suppose the individual has labor income of zero in the first period and Y_2 in the second. Second-period consumption is thus $Y_2 - (1+r)C_1$. Y_2 is certain; again, r may be random.

(i) Find the first-order condition for the individual's choice of C_1.

(ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?

Problem 5.7

(a) Use an argument analogous to that used to derive equation (5.23) to show that household optimization requires

$$\frac{b}{1-\ell_t} = e^{-\rho}\,\mathrm{E}_t\left[\frac{b\,w_t\,(1+r_{t+1})}{w_{t+1}\,(1-\ell_{t+1})}\right]$$

(b) Show that this condition is implied by (5.23) and (5.26). (Note that [5.26] must hold in every period.)

Problem 5.8 *A simplified real-business-cycle model with additive technology shocks.*

(Blanchard and Fischer, 1989, pp. 329–331) Consider an economy consisting of a constant population of infinitely lived individuals. The representative individual maximizes the expected value of $\sum_{t=0}^{\infty} u(C_t)/(1+\rho)^t$, $\rho > 0$. The instantaneous utility function, $u(C_t)$, is $u(C_t) = C_t - \theta C_t^2$, $\theta > 0$. Assume that C is always in the range where $u'(C)$ is positive.

Output is linear in capital, plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation; thus $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A. Assume $A = \rho$. Finally, the disturbance follows a first-order autoregressive process: $e_t = \phi e_{t-1} + \varepsilon_t$, where $-1 < \phi < 1$ and where the ε_t's are mean-zero, i.i.d. shocks.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

(b) Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma e_t$. Given this guess, what is K_{t+1} as a function of K_t and e_t?

(c) What values must the parameters α, β, and γ have for the first-order condition in part (a) to be satisfied for all values of K_t and e_t?

(d) What are the effects of a one-time shock to ε on the paths of Y, K, and C?

Problem 5.9 *A simplified real-business-cycle model with taste shocks.*

(Blanchard and Fischer, 1989, p. 361) Consider the setup in Problem 5.8. Assume, however, that the technological disturbances (the e's) are absent and that the instantaneous utility function is $u(C_t) = C_t - \theta(C_t + \nu_t)^2$. The ν's are mean-zero, i.i.d. shocks.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

(b) Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma \nu_t$. Given this guess, what is K_{t+1} as a function of K_t and ν_t?

(c) What values must the parameters α, β, and γ have for the first-order condition in (a) to be satisfied for all values of K_t and ν_t?

(d) What are the effects of a one-time shock to ν on the paths of Y, K, and C?

Problem 5.10 *The balanced growth path of the model of Section 5.3.*

Consider the model of Section 5.3 without any shocks. Let y^*, k^*, c^*, and G^* denote the values of $Y/(AL)$, $K/(AL)$, $C/(AL)$, and $G/(AL)$ on the balanced growth path; w^* the value of w/A; ℓ^* the value of L/N; and r^* the value of r.

(a) Use equations (5.1)–(5.4), (5.23), and (5.26) and the fact that y^*, k^*, c^*, w^*, ℓ^*, and r^* are constant on the balanced growth path to find six equations in these six variables.

Hint: The fact that c in [5.23] is consumption per person, C/N, and c^* is the balanced-growth-path value of consumption per unit of effective labor, $C/(AL)$, implies that $c = c^*\ell^*A$ on the balanced growth path.

(b) Consider the parameter values assumed in Section 5.7. What are the implied shares of consumption and investment in output on the balanced growth path? What is the implied ratio of capital to annual output on the balanced growth path?

Problem 5.11 *Solving a real-business-cycle model by finding the social optimum.*

Consider the model of Section 5.5. Assume for simplicity that $n = g = \bar{A} = \bar{N} = 0$. Let $V(K_t, A_t)$, the *value function*, be the expected present value from the current period forward of lifetime utility of the representative individual as a function of the capital stock and technology.

(a) Explain intuitively why $V(\cdot)$ must satisfy

$$V(K_t, A_t) = \max_{C_t, \ell_t} \{[\ln C_t + b\ln(1-\ell_t)] + e^{-\rho} E_t[V(K_{t+1}, A_{t+1})]\}.$$

This condition is known as the *Bellman equation*.

Given the log-linear structure of the model, let us guess that $V(\cdot)$ takes the form $V(K_t, A_t) = \beta_0 + \beta_K \ln K_t + \beta_A \ln A_t$, where the values of the β's are to be determined. Substituting this conjectured form and the facts that $K_{t+1} = Y_t - C_t$ and $E_t[\ln A_{t+1}] = \rho_A \ln A_t$ into the Bellman equation yields

$$V(K_t, A_t) = \max_{C_t, \ell_t} \left\{[\ln C_t + b\ln(1-\ell_t)] + e^{-\rho}\left[\beta_0 + \beta_K \ln(Y_t - C_t) + \beta_A \rho_A \ln A_t\right]\right\}.$$

(b) Find the first-order condition for C_t. Show that it implies that C_t/Y_t does not depend on K_t or A_t.

(c) Find the first-order condition for ℓ_t. Use this condition and the result in part (b) to show that ℓ_t does not depend on K_t or A_t.

(d) Substitute the production function and the results in parts (b) and (c) for the optimal C_t and ℓ_t into the equation above for $V(\cdot)$, and show that the resulting expression has the form $V(K_t, A_t) = \beta'_0 + \beta'_K \ln K_t + \beta'_A \ln A_t$.

(e) What must β_K and β_A be so that $\beta'_K = \beta_K$ and $\beta'_A = \beta_A$?

(f) What are the implied values of C/Y and ℓ? Are they the same as those found in Section 5.5 for the case of $n = g = 0$?

Problem 5.12

Suppose technology follows some process other than (5.8)–(5.9). Do $s_t = \hat{s}$ and $\ell_t = \hat{\ell}$ for all t continue to solve the model of Section 5.5? Why or why not?

Problem 5.13

Consider the model of Section 5.5. Suppose, however, that the instantaneous utility function, u_t, is given by $u_t = \ln c_t + b(1-\ell_t)^{1-\gamma}/(1-\gamma)$, $b > 0$, $\gamma > 0$, rather than by (5.7) (see Problem 5.4).

(a) Find the first-order condition analogous to equation (5.26) that relates current leisure and consumption, given the wage.

(b) With this change in the model, is the saving rate (s) still constant?

(c) Is leisure per person $(1-\ell)$ still constant?

Problem 5.14

(a) If the $\tilde{A}_t$'s are uniformly 0 and if $\ln Y_t$ evolves according to (5.39), what path does $\ln Y_t$ settle down to?

Hint: Note that we can rewrite [5.39] as

$$\ln Y_t - (n+g)\,t = Q + \alpha\left[\ln Y_{t-1} - (n+g)(t-1)\right] + (1-\alpha)\tilde{A}_t$$

where $Q \equiv \alpha \ln \hat{s} + (1-\alpha)\left(\bar{A} + \ln \hat{\ell} + \bar{N}\right) - \alpha(n+g)$.

(b) Let $\tilde{Y}_t$ denote the difference between $\ln Y_t$ and the path found in (a). With this definition, derive (5.40).

Problem 5.15 *The derivation of the log-linearized equation of motion for capital.*

Consider the equation of motion for capital, $K_{t+1} = K_t + K_t^{\alpha}(A_t L_t)^{1-\alpha} - C_t - G_t - \delta K_t$.

(a) (i) Show that

$$\frac{\partial \ln K_{t+1}}{\partial \ln K_t} = \frac{1+r_t}{K_{t+1}/K_t}$$

(ii) Show that this implies that $\partial \ln K_{t+1}/\partial \ln K_t$ evaluated at the balanced growth path is $(1+r^*)/e^{n+g}$.

(b) Show that

$$\tilde{K}_{t+1} \simeq \lambda_1 \tilde{K}_t + \lambda_2 \left(\tilde{A}_t + \tilde{L}_t\right) + \lambda_3 \tilde{G}_t + (1-\lambda_1-\lambda_2-\lambda_3)\,\tilde{C}_t,$$

where

$$\lambda_1 \equiv \frac{(1+r^*)}{e^{n+g}}, \quad \lambda_2 \equiv \frac{(1-\alpha)(r^*+\delta)}{\alpha\, e^{n+g}}, \quad \lambda_3 \equiv -\frac{(r^*+\delta)(G/Y)^*}{\alpha\, e^{n+g}}$$

and where $(G/Y)^*$ denotes the ratio of G to Y on the balanced growth path without shocks. (Hints: Since the production function is Cobb–Douglas, $Y^* = (r^*+\delta)K^*/\alpha$. On the balanced growth path, $K_{t+1} = e^{n+g}K_t$, which implies that $C^* = Y^* - G^* - \delta K^* - (e^{n+g}-1)K^*$.)

(c) Use the result in (b) and equations (5.43)–(5.44) to derive (5.53), where $b_{KK} = \lambda_1 + \lambda_2 a_{LK} + (1-\lambda_1-\lambda_2-\lambda_3)\,a_{CK}$, $b_{KA} = \lambda_2(1+a_{LA}) + (1-\lambda_1-\lambda_2-\lambda_3)\,a_{CA}$, and $b_{KG} = \lambda_2 a_{LG} + \lambda_3 + (1-\lambda_1-\lambda_2-\lambda_3)\,a_{CG}$.

Problem 5.16

Redo the regression reported in equation (5.55):

(a) Incorporating more recent data.

(b) Incorporating more recent data, and using $M1$ rather than $M2$.

(c) Including eight lags of the change in log money rather than four.

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